

HW 6.8.1.c: Converting Polar and Rectangular Equations

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$r^2 = x^2 + y^2$$

$$\tan \theta = \frac{y}{x}$$

Using the equations above, convert the given Cartesian equation to a polar equation.

1. $x = 3$

2. $y = -1$

3. $x^2 + y^2 = 16$

4. $(x-6)^2 + y^2 = 36$

5. $x^2 + y^2 = 4y$

6. $x^2 + y^2 = 5x$

Convert the given polar equation to a Cartesian equation.

7. $r = 3 \sin(\theta)$

8. $r = 5 \cos(\theta)$

9. $r = 2 \sec(\theta)$

10. $r = -2 \csc(\theta)$

11. $r = -5 \sin(\theta)$

12. $r = 7$

Selected Answers:

1. $r = 3 \sec(\theta)$

3. $r = 4$

5. $r = 4 \sin(\theta)$

7. $x^2 + \left(y - \frac{3}{2}\right)^2 = \frac{9}{4}$

9. $x = 2$

11. $x^2 + (y + 2.5)^2 = 6.25$